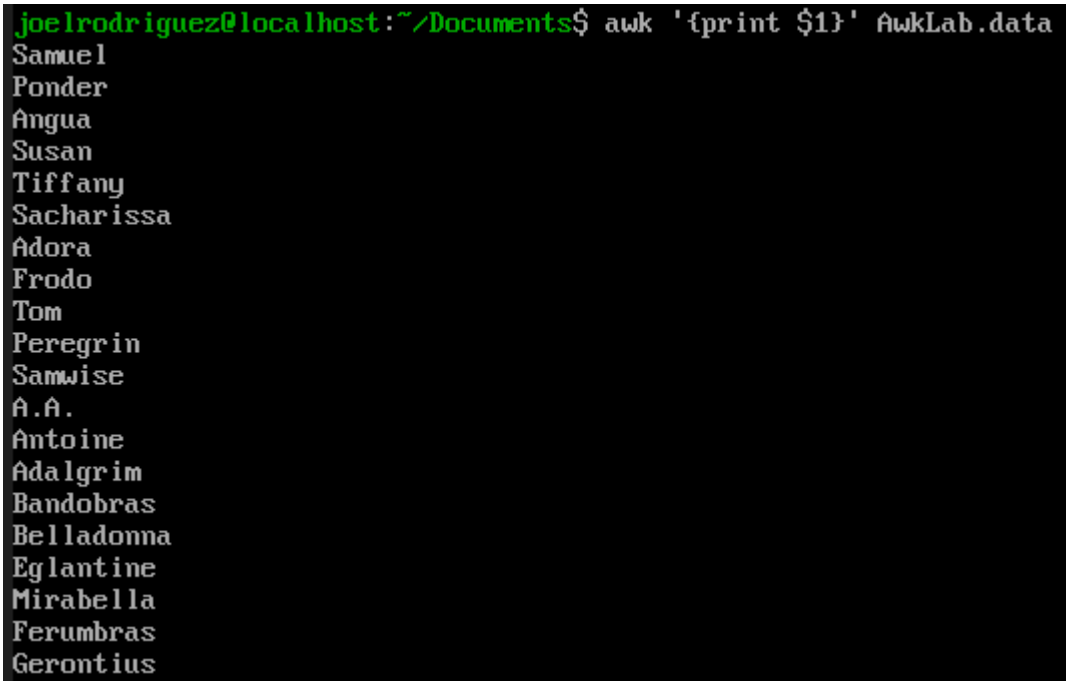
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Fall 2026
CIS 245
Awk Lab

1. Print all the First Names. - **COMMAND**: awk ‘{print $1}’ AwkLab.data



Source: https://www.youtube.com/watch?v=4HpWO_RAMq4&t=57s

This command works because AWK separates fields by spaces by default, $1 selects the first name, and it will be displayed when printed

2. Print phone numbers for Tom and Frodo after their names
COMMAND: awk -F: ‘/Tom/ || /Frodo/ “ {print $1 “\t” $2}’ AwkLab.data



This command works because I used -F: to separate the fields using a colon, “$1” to print their full names, and “$2” prints their phone numbers.

3. Print Peregrin's full name and phone number area code only.
   COMMAND: awk '/Peregrin/ {print $1, $2} AwkLad.data

```
joelrodriguez@localhost:~/Documents$ awk '/Peregrin/ {print $1, $2}' AwkLab.data
Peregrin Took:(510)
```

   Source: https://www.grymoire.com/Unix/Awk.html

   This command works because once AWK find the matching record, "$1" prints his full name and $2 prints his area code.

4. Print all phone numbers (full number) in the 123 area code along with the names
   COMMAND: awk -F: '/123/ {print $1 "\t" $2}' Awklab.data

```
joelrodriguez@localhost:~/Documents$ awk -F: '/123/ {print $1 "\t" $2}' AwkLab.data
Antoine de Saint-Exupery        (123) 978-6432
Belladonna Took (123) 978-5754
Eglantine Took  (123) 978-3574
```

   Source: https://www.youtube.com/watch?v=4HpWO_RAMq4&t=57s

   This command works because -F: separates the fields using a colon, and /123/ searches for records containing that area code. Once Awk finds a matching record, "$1" prints the person's full name and "$2" print their full phone number. I used \t to add a tab between the name and phone number so the output is easier to read.

5. Print all Last names beginning with either a T or D (careful of middle names!) COMMAND: ./5question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./5question AwkLab.data
Dearheart
Took
Took
Took
Took
Took
Took
Took
Took
joelrodriguez@localhost:~/Documents$ cat 5question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{
        n = split($1, name, " ")

        if (name[n] ~ /[TD]/)
        {
                print name[n]
        }
}
```

Source: https://www.grymoire.com/Unix/Awk.html

This script works because FS separates the fields using a colon, and the split() separates the full name into individual words. I used n to store the number of words in the name and name [n] to selects the last word, so middle names don't interfere. The ~ checks for a match, [TD] checks whether it starts with T or D. The AWK will only print if the condition is true.

6. Print all first names containing four or less characters.
   COMMAND: ./6question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./6question AwkLab.data
Tom
A.A.
joelrodriguez@localhost:~/Documents$ cat 6question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

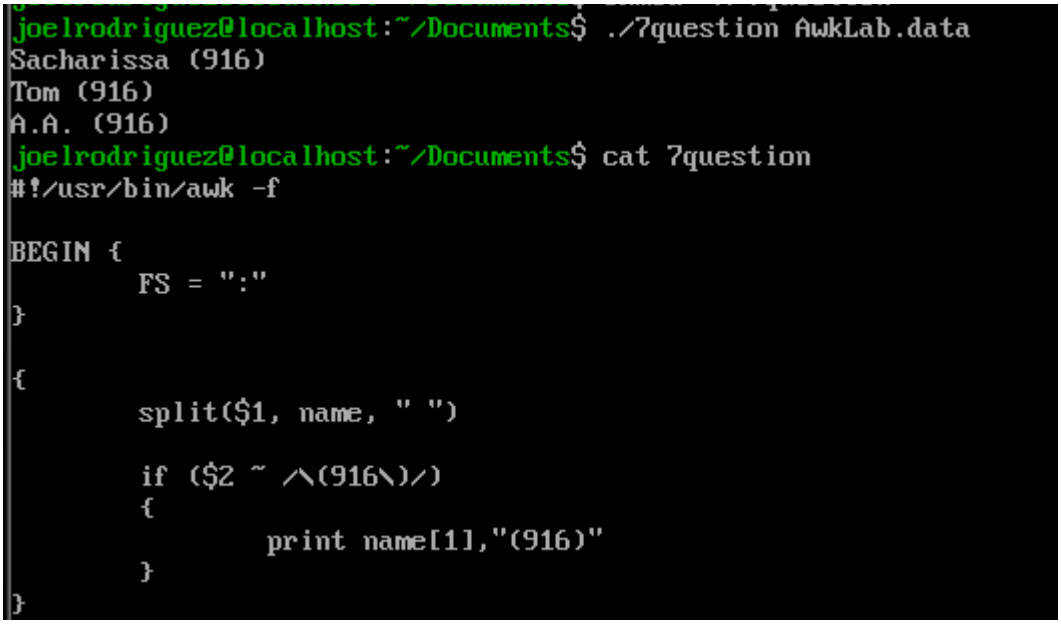
{
        n = split($1, name, " ")

        if (length(name[1]) <=4)
        {
                print name[1]
        }
}
```

   Source: https://www.grymoire.com/Unix/Awk.html

   This script works because the FS seperates the fields using a colon and split() seperates the full name into individual words. I used name [1] to select the first name and the length() function counts the characters it contains. The “<=4” checks if the word has four or fewer letters. If the condtion is true, AWK will print the person’s first name,

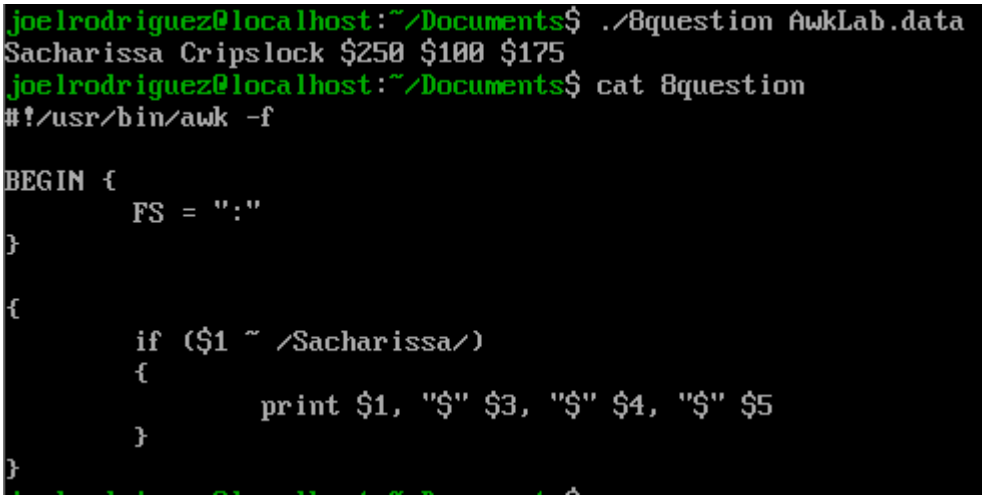
7. Print the first names and area codes of all those in the 916 area code.
   COMMAND: ./7question AwkLab.data



   Source: https://www.grymoire.com/Unix/Awk.html

   This script works because the FS seperates the fields by colons. Split() seperates the full name into individual words and name[1] to select the first name. The IF statement checks whether the phone number in $2 contains the 916 area code and if it does, AWK will print the person's first name and area code.

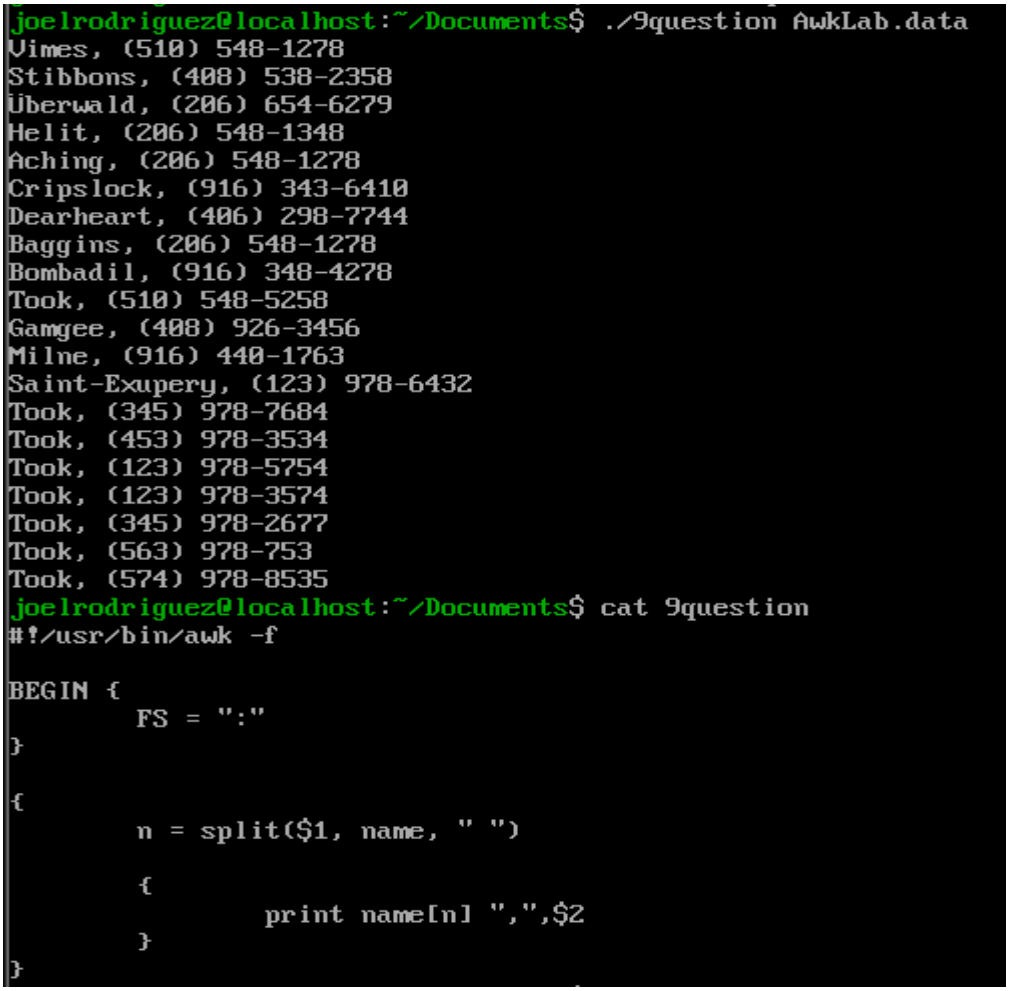
8. Print Sacharissa's campaign contributions following her name. Each value should be printed with a leading dollar sign; e.g., \$250 \$100 \$175.
   COMMAND: ./8question AwkLab.data



   Source: https://www.grymoire.com/Unix/Awk.html

   This command works because the FS separates the fields using a colon, and the IF statement checks whether the first field contains Sacharissa's name. I used $1 to print her full name and $3 $4 $5 to print her contributions. The dollar signs displays the amount as money.

9. Print last names followed by a comma and the phone number. Be careful of the last names's format. - COMMAND: ./9question AwkLab.data



```
joelrodriguez@localhost:~/Documents$ ./9question AwkLab.data
Vimes, (510) 548-1278
Stibbons, (408) 538-2358
Überwald, (206) 654-6279
Helit, (206) 548-1348
Aching, (206) 548-1278
Cripslock, (916) 343-6410
Dearheart, (406) 298-7744
Baggins, (206) 548-1278
Bombadil, (916) 348-4278
Took, (510) 548-5258
Gamgee, (408) 926-3456
Milne, (916) 440-1763
Saint-Exupery, (123) 978-6432
Took, (345) 978-7684
Took, (453) 978-3534
Took, (123) 978-5754
Took, (123) 978-3574
Took, (345) 978-2677
Took, (563) 978-753
Took, (574) 978-8535
joelrodriguez@localhost:~/Documents$ cat 9question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{
        n = split($1, name, " ")

        {
                print name[n] ",",$2
        }
}
```

Source: https://www.grymoire.com/Unix/Awk.html

This script works because FS separated the field using colon, and Split () sepearates the full name into individual words. I used n to store the number of words and name[n] to select the last word. Print will display the last word, a comma, and the phone number from $2.

10. Print the first and last names of those who contributed more than \$110 in the last month. Make sure to include their last month contribution amount after the name COMMAND: ./10question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./10question AwkLab.data
Samuel Vimes $175
Ponder Stibbons $201
Susan Helit $175
Tiffany Aching $150
Sacharissa Cripslock $175
Adora Dearheart $275
Tom Bombadil $175
Peregrin Took $135
Samwise Gamgee $200
A.A. Milne $300
Antoine Saint-Exupery $175
Adalgrim Took $467
Bandobras Took $4673
Belladonna Took $175
Eglantine Took $4367
Mirabella Took $175
Ferumbras Took $3457
Gerontius Took $4562
joelrodriguez@localhost:~/Documents$ cat 10question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{
        n = split($1, name, " ")

        if ($5 > 110)
        {
                print name[1], name[n], "$" $5
        }
}
```

Source: https://www.grymoire.com/Unix/Awk.html

This script works because FS separated the field using colon, and Split () separates the full name into individual words. I used name[1] to select the first name and name [n] to select the last name. The IF statement checks

whether the contribution in $5 is greater than $110. If the condition is true, AWK will the person's first and last name and last month's contribution.

11. Print the last names, phone numbers, and first month contribution of those who contributed less than \$150 in the first month.
    COMMAND: ./11question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./11question AwkLab.data
Aching (206) 548-1278 $15
Took (510) 548-5258 $50
joelrodriguez@localhost:~/Documents$ cat 11question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{
        n = split($1, name, " ")

        if ($3 < 150)
        {
                print name[n], $2, "$" $3
        }
}
```

    Source: https://www.grymoire.com/Unix/Awk.html
    This script works because FS separated the field using colon, and Split () separates the full name into individual words. I used n to store the number of words and name[n] to select the last word. The IF statement checks whether the first month contribution in "$3" is less than $150. AWK will print the person's last name, phone number and the first month's contribution with a dollar sign.

12. Print the first names and contribution of those who contributed between \$10 and \$200 in the first month. - COMMAND: ./12question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./12question AwkLab.data
Ponder $155
Tiffany $15
Peregrin $50
A.A. $175
joelrodriguez@localhost:~/Documents$ cat 12question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{
        split($1, name, " ")

        if ($3 >= 10 && $3<=200)
        {
                print name[1], "$" $3
        }
}
```

Source: https://www.grymoire.com/Unix/Awk.html

This script works because FS separated the field using colon, and Split () separates the full name into individual words. I used name[1] to select the first name and $3 to check the first month contribution. The && makes sure the contribution is greater than or equal to $10 or less than or equal to $200. If both conditions are true, AWK will print the person's first name and contribution with a dollar sign.

13. Print the first name, last names and total contributions of those who contributed less than \$700 over the three-month period. - COMMAND: ./13question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./13question AwkLab.data
Samuel Vimes $525
Ponder Stibbons $446
Angua Überwald $360
Susan Helit $525
Tiffany Aching $353
Sacharissa Cripslock $525
Frodo Baggins $405
Tom Bombadil $525
Peregrin Took $280
Samwise Gamgee $618
A.A. Milne $550
Antoine Saint-Exupery $525
joelrodriguez@localhost:~/Documents$ cat 13question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{
        n = split($1, name, " ")

        Total = $3 + $4 + $5

        if (Total < 700)
        {
                print name[1], name[n], "$" Total
        }
}
```

Source: https://www.grymoire.com/Unix/Awk.html

This script works because FS separated the field using colon, and Split () separates the full name into individual words. I used n to store the number of words, used name[1] to select the first name and name[n] to select the last name. The Total variable adds the contributions from $3 $4 and $5. The IF statement checks whether the total is less than $700. AWK will print the person's first name, last name and total contribution with a dollar sign if the condition is true.

14. Print the first names and first letter of the last name, and average contribution of those who had an average contribution of more then \$300
COMMAND: ./14question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./14question AwkLab.data
Adora D $341.667
Adalgrim T $1746.67
Bandobras T $3931.33
Eglantine T $1771.67
Mirabella T $507
Ferumbras T $1269
Gerontius T $1925
joelrodriguez@localhost:~/Documents$ cat 14question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{
        n = split($1, name, " ")

        Average = ($3 + $4 + $5) / 3

        if (Average > 300)
        {
                print name[1], substr(name[n], 1, 1), "$" Average
        }
}
```

Source: https://www.grymoire.com/Unix/Awk.html

This script works because FS separated the field using colon, and Split () separates the full name into individual words. I used name[1] to select the first name and name[n] to select the last name. The Average variable adds the three monthly contributions and divides the total by three. The IF statement checks whether average is greater than $300. If the condition is true, AWK uses substr() to get the first letter of the last name and prints it along with the first name and average contribution.

15. Print the last name and area code of those not in the 916 area code.
    COMMAND: ./15question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./15question AwkLab.data
Vimes (510)
Stibbons (408)
Überwald (206)
Helit (206)
Aching (206)
Dearheart (406)
Baggins (206)
Took (510)
Gamgee (408)
Saint-Exupery (123)
Took (345)
Took (453)
Took (123)
Took (123)
Took (345)
Took (563)
Took (574)
joelrodriguez@localhost:~/Documents$ cat 15question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{
        n = split($1, name, " ")

        if ($2 !~ /\(916\)/)
        {
                print name[n], substr($2, 1, 5)
        }
}
joelrodriguez@localhost:~/Documents$
```

    Source: https://www.grymoire.com/Unix/Awk.html

    This script works because FS separated the field using colon, and Split () separates the full name into individual words. I used n to store the number of words and name[n] to select the last name. The !~ checks whether the phone number is $2 does not contain the 916 area code. If the condition is true, AWK uses substr() extract the area code and prints it alongside the person's last name.

16. Print each record preceded by the number of the record. -

    COMMAND: awk {print NR, $0} AwkLab.data

```
joelrodriguez@localhost:~/Documents$ awk '{print NR, $0}' AwkLab.data
1 Samuel Vimes:(510) 548-1278:250:100:175
2 Ponder Stibbons:(408) 538-2358:155:90:201
3 Angua von Überwald:(206) 654-6279:250:60:50
4 Susan Sto Helit:(206) 548-1348:250:100:175
5 Tiffany Aching:(206) 548-1278:15:188:150
6 Sacharissa Cripslock:(916) 343-6410:250:100:175
7 Adora Belle Dearheart:(406) 298-7744:450:300:275
8 Frodo Baggins:(206) 548-1278:250:80:75
9 Tom Bombadil:(916) 348-4278:250:100:175
10 Peregrin Took:(510) 548-5258:50:95:135
11 Samwise Gamgee:(408) 926-3456:250:168:200
12 A.A. Milne:(916) 440-1763:175:75:300
13 Antoine de Saint-Exupery:(123) 978-6432:250:100:175
14 Adalgrim Took:(345) 978-7684:4673:100:467
15 Bandobras "Bullroarer" Took:(453) 978-3534:6753:368:4673
16 Belladonna Took:(123) 978-5754:356:247:175
17 Eglantine Took:(123) 978-3574:473:475:4367
18 Mirabella Took:(345) 978-2677:783:563:175
19 Ferumbras III Took:(563) 978-753:250:100:3457
20 Gerontius Took:(574) 978-8535:535:678:4562
```

    Source: https://www.grymoire.com/Unix/Awk.html

    This command works because NR automatically keeps track of the record number, while the “$0” represents the entire line. This command will display the record number followed by the complete record.

17. Print the name and total contribution of each person.

    COMMAND: ./17question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./17question AwkLab.data
Samuel Vimes $525
Ponder Stibbons $446
Angua von Überwald $360
Susan Sto Helit $525
Tiffany Aching $353
Sacharissa Cripslock $525
Adora Belle Dearheart $1025
Frodo Baggins $405
Tom Bombadil $525
Peregrin Took $280
Samwise Gamgee $618
A.A. Milne $550
Antoine de Saint-Exupery $525
Adalgrim Took $5240
Bandobras "Bullroarer" Took $11794
Belladonna Took $778
Eglantine Took $5315
Mirabella Took $1521
Ferumbras III Took $3807
Gerontius Took $5775
joelrodriguez@localhost:~/Documents$ cat 17question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{
        Total = $3 + $4 + $5
        {
                print $1, "$" Total
        }
}
```

    Source: https://www.grymoire.com/Unix/Awk.html

    This script works because FS separated the field using colon, and “$1” contains the person’s full name. I sued the Total variable to add the three monthly contributions from “$3”, “$4”, and “$5”. The AWK will print each person’s full name followed by their total contribution with a dollar sign.

18. Add \$10 to Tiffany Aching's first contribution and print her full name and first contribution. - COMMAND: ./18question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./18question AwkLab.data
Tiffany Aching $25
joelrodriguez@localhost:~/Documents$ cat 18question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{

        if ($1 == "Tiffany Aching")
        {
                $3 = $3 + 10
                print $1, "$" $3
        }
}
```

Source: https://www.grymoire.com/Unix/Awk.html

This script works because FS separated the field using colon and the IF statement checks whether “$1” matches her full name. I used $3 = $3 + 10 to increase her first contribution by ten dollars. AWK will print her first name and updated contribution with a dollar sign.

19. Change Samwise Gamgee's name to Sean Astin -
    COMMAND: ./19question AwkLab.data

```
joelrodriguez@localhost:~/Documents$ ./19question AwkLab.data
Samuel Vimes:(510) 548-1278:250:100:175
Ponder Stibbons:(408) 538-2358:155:90:201
Angua von Überwald:(206) 654-6279:250:60:50
Susan Sto Helit:(206) 548-1348:250:100:175
Tiffany Aching:(206) 548-1278:15:188:150
Sacharissa Cripslock:(916) 343-6410:250:100:175
Adora Belle Dearheart:(406) 298-7744:450:300:275
Frodo Baggins:(206) 548-1278:250:80:75
Tom Bombadil:(916) 348-4278:250:100:175
Peregrin Took:(510) 548-5258:50:95:135
Sean Astin:(408) 926-3456:250:168:200
A.A. Milne:(916) 440-1763:175:75:300
Antoine de Saint-Exupery:(123) 978-6432:250:100:175
Adalgrim Took:(345) 978-7684:4673:100:467
Bandobras "Bullroarer" Took:(453) 978-3534:6753:368:4673
Belladonna Took:(123) 978-5754:356:247:175
Eglantine Took:(123) 978-3574:473:475:4367
Mirabella Took:(345) 978-2677:783:563:175
Ferumbras III Took:(563) 978-753:250:100:3457
Gerontius Took:(574) 978-8535:535:678:4562
joelrodriguez@localhost:~/Documents$ cat 19question
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
        OFS = ":"
}

{
        if ($1 == "Samwise Gamgee")
            $1 = "Sean Astin"
        {
                print $0
        }
}
```

Source: https://www.grymoire.com/Unix/Awk.html
This script works because FS separated the field using colon, while the OFS keeps the same separator when printing the new updated record. The IF statement checks whether “$1” matches Samwise Gamgee’s full name and if it does AWK will replace it with Sean Astin. “$0” prints the entire record with the updated name.

20. Write an awk script to do the following (MUST be an awk script not just a bash script or commands on the commandline) enumerate
21. Prints first name of the all the Tooks followed by their total campaign contributions .

    The FS separates the field using a colon, and split() separates each person's full name into individual words. I used name[n] to check whether the last name is Took. The total variable adds the contributions from all three months. If the last name matches, AWK prints the person's first name and total contribution.
22. Print the full names and contributions of anyone who contributed between \$10 and \$200 in the last contribution

    The "$5" field contains the last month contribution and && checks that the amount is greater than or equal to $10 and less than or equal to $200. If both conditions are true, AWK will print the person's full name from "$1" and their last month contribution.
23. Prints the full names and average contribution of those who contributed less than \$300 on average

    The total variable adds the contributions from "$3", "$4", and "$5", and average divides the total by three. The IF statement checks whetherthe average is less than $300. If the condition is true, AWK will print the Person's full name and average contribution.

    Screenshots are below. The outputs are mixed together but the label of what the question it goes to is next to them.

```
joelrodriguez@localhost:~/Documents$ ./finalquestion AwkLab.data
Q22: Samuel Vimes $175
Q23: Samuel Vimes $175
Q23: Ponder Stibbons $148.667
Q22: Angua von Überwald $50
Q23: Angua von Überwald $120
Q22: Susan Sto Helit $175
Q23: Susan Sto Helit $175
Q22: Tiffany Aching $150
Q23: Tiffany Aching $117.667
Q22: Sacharissa Cripslock $175
Q23: Sacharissa Cripslock $175
Q22: Frodo Baggins $75
Q23: Frodo Baggins $135
Q22: Tom Bombadil $175
Q23: Tom Bombadil $175
Q21: Peregrin $280
Q22: Peregrin Took $135
Q23: Peregrin Took $93.3333
Q22: Samwise Gamgee $200
Q23: Samwise Gamgee $206
Q23: A.A. Milne $183.333
Q22: Antoine de Saint-Exupery $175
Q23: Antoine de Saint-Exupery $175
Q21: Adalgrim $5240
Q21: Bandobras $11794
Q21: Belladonna $778
Q22: Belladonna Took $175
Q23: Belladonna Took $259.333
Q21: Eglantine $5315
Q21: Mirabella $1521
Q22: Mirabella Took $175
Q21: Ferumbras $3807
Q21: Gerontius $5775
```

```
joelrodriguez@localhost:~/Documents$ cat finalquestion
#!/usr/bin/awk -f

BEGIN {
        FS = ":"
}

{
        n = split($1, name, " ")
        total = $3 + $4 +$5
        average = total / 3

        #Question 21
        if (name[n] == "Took")
        {

                print "Q21:", name[1], "$" total
        }

        #Question 22
        if ($5 >= 10 && $5 <= 200)
        {
                print "Q22:", $1, "$" $5
        }

        #Question 23
        if (average < 300)
        {
                print "Q23:", $1, "$" average
        }
}
```